



GTC4Lusso USA - Engine oil pump - Latest Update: 2021-01-26

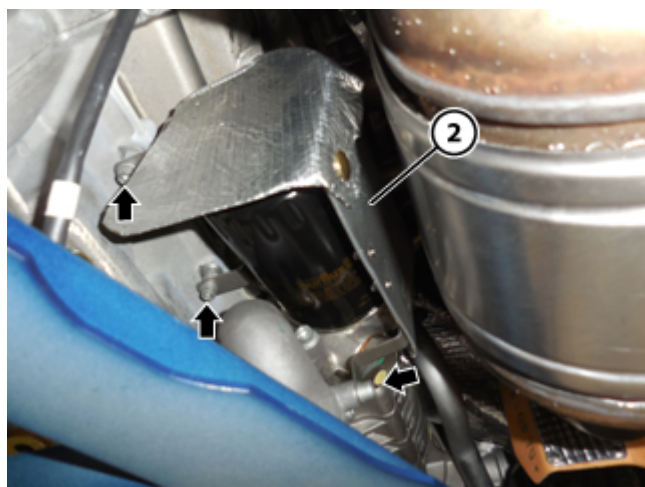
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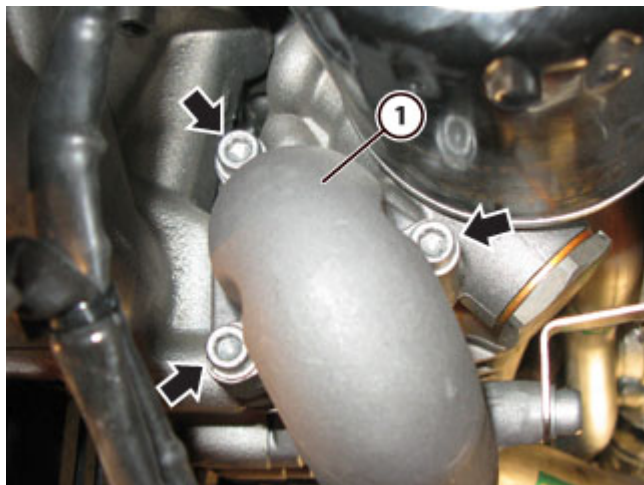
B5.03 Engine oil pump

Removing the water/oil pump

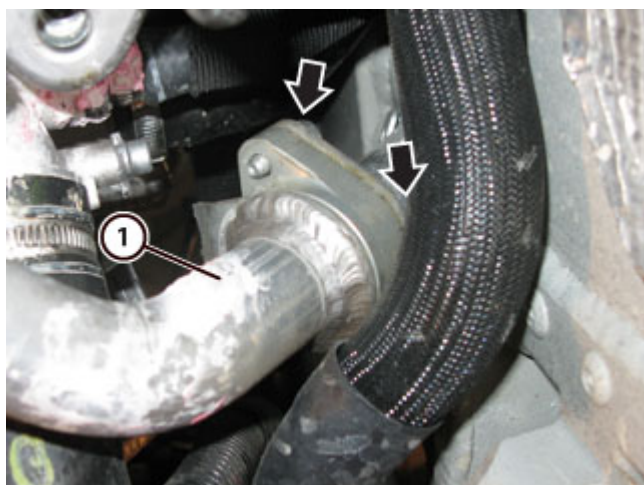
- Remove the front wheelhouses (🔧 E3.05).
 - ❗ Remove the front right hand wheelhouse.
- Drain the engine cooling system (🔧 B6.07).
- Drain the engine lubrication system (🔧 A3.01).
- Remove the front removable subframe (🔧 E2.02).
- Remove the catalytic converters (🔧 B7.03).
 - ❗ Remove the right hand catalytic converter.



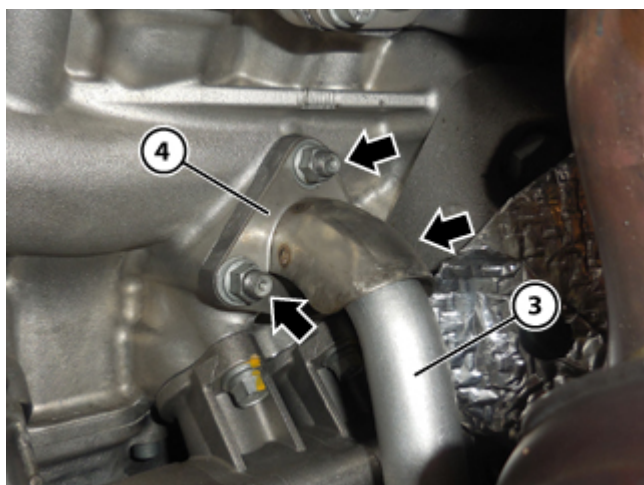
- Remove the heat shield (2), by undoing the screws indicated.



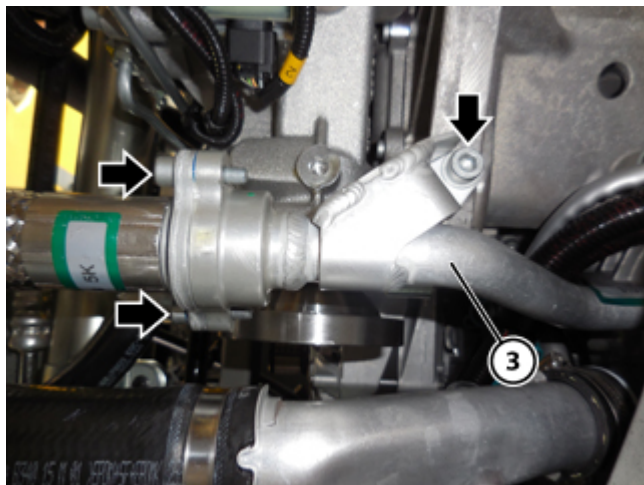
- Undo the indicated nuts on the engine oil pipe (1).



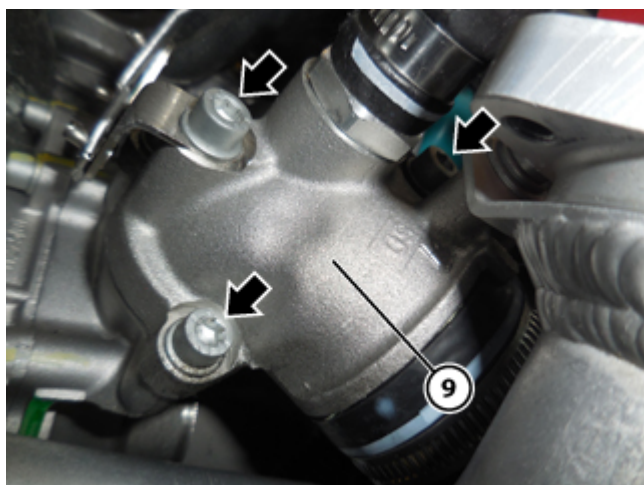
- Undo the indicated screws and remove the engine oil pipe (1).



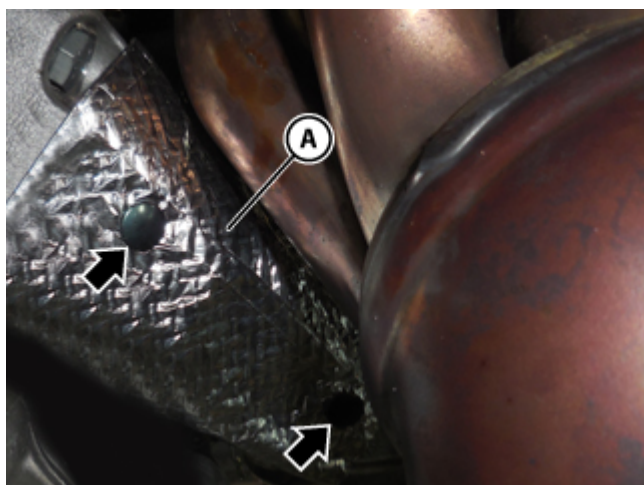
- Undo the indicated nuts fastening the engine oil pipe (3) to the crankcase and retrieve the shield (4).



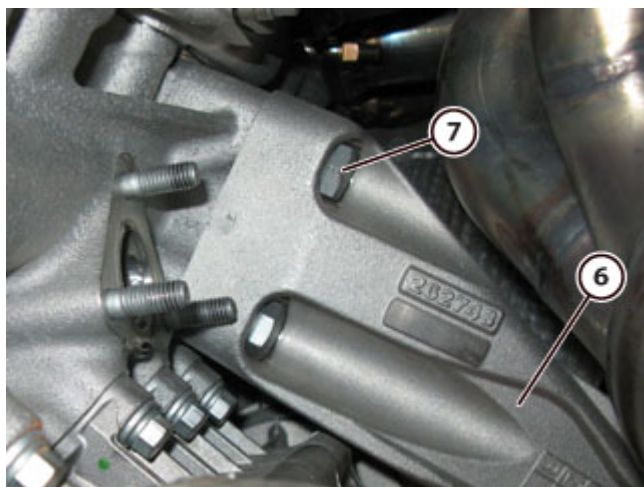
- Undo the indicated screw.
- Remove the engine oil pipe (3), undoing the indicated screws.



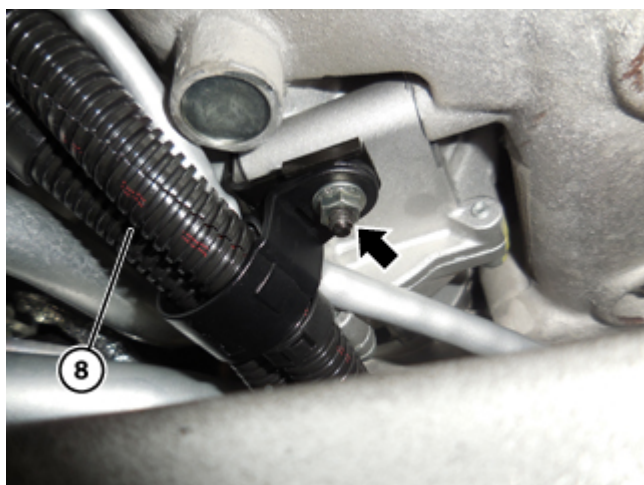
- Remove the hose union (9) from the pump, undoing the indicated screws.



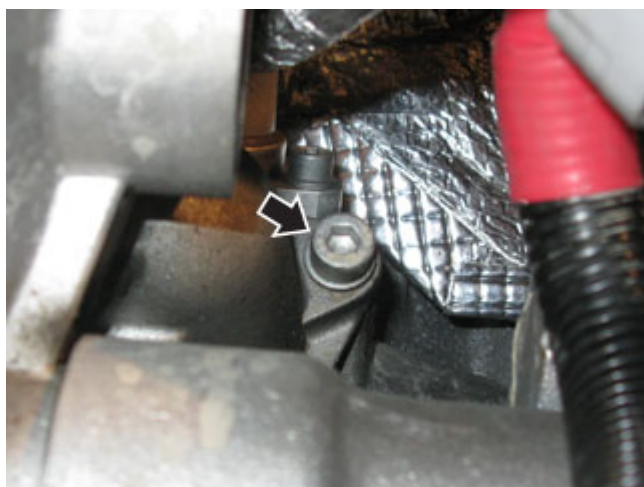
- Remove the heat shield (A), releasing the indicated clips.



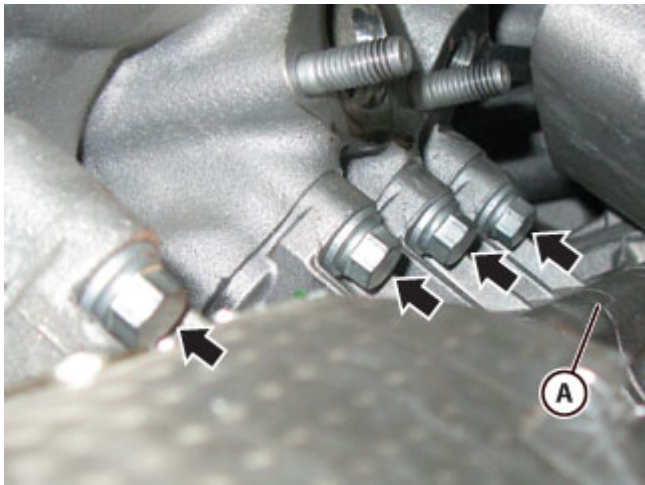
- Undo the four screws (7) and move the engine mounting strut (6) aside.



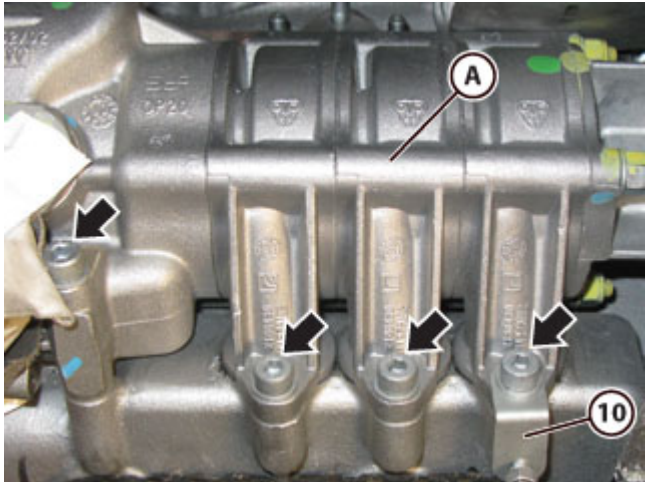
- Undo the indicated screw to release the engine services cable (8).



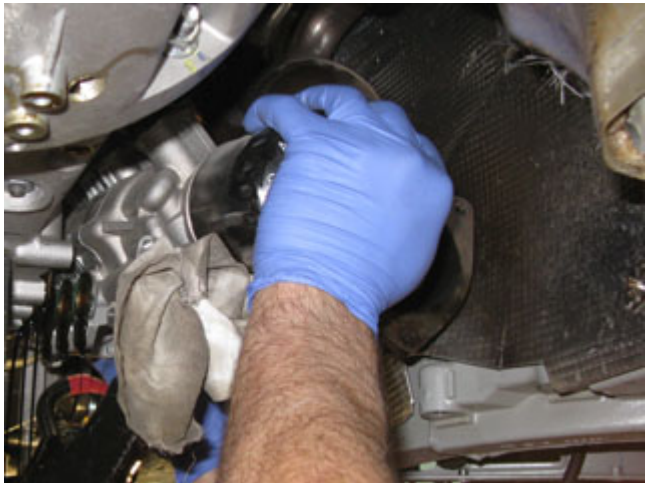
- Undo the indicated screw.



- Undo the indicated screws fastening the water/oil pump (A).



- Undo the indicated screws fastening the water/oil pump (A).
- Release the bracket (10).

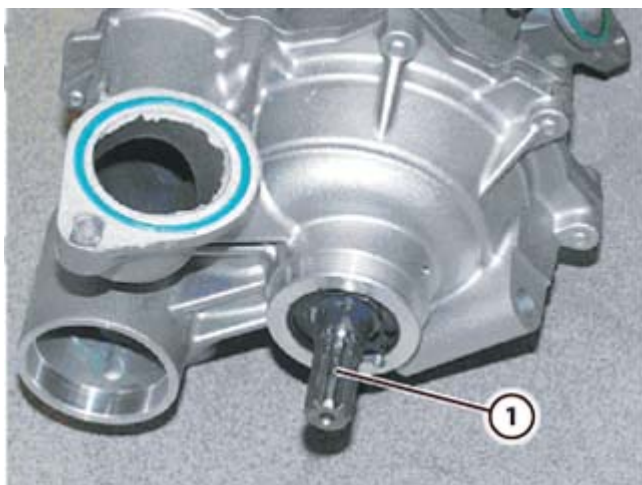


- Ease out the water/oil pump horizontally towards the rear and remove.

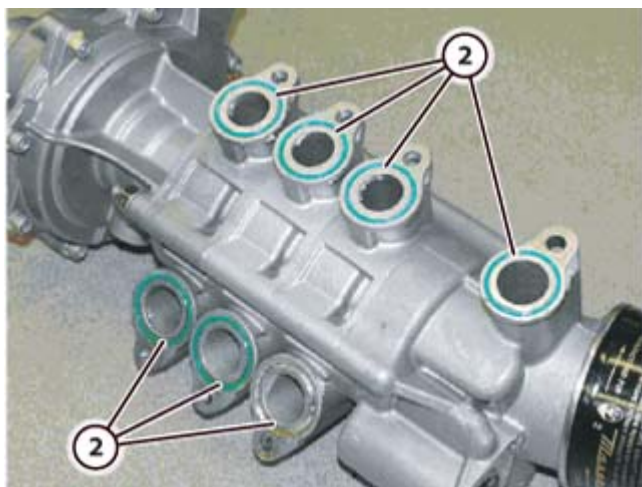
Refitting the water/oil pump

			
Tightening torque		Nm	Class
Fastening water/oil pump to crankcase	Screw (pretightening)	5 Nm	0
	Screw	25 Nm	B

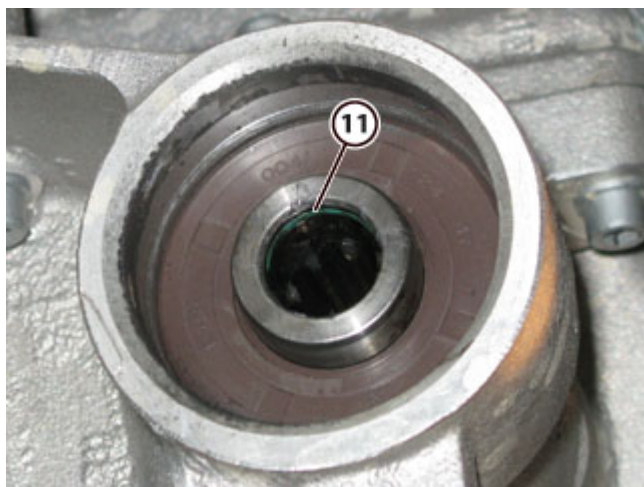
			
Tightening torque		Nm	Class
Fastening engine mounting strut to crankcase	Screw	45 Nm	A
Fastening hose connector to water/oil pump	Screw	10 Nm	B
Fastening oil pipe leading from engine to disconnecter	Nut	20 Nm	B
Fastening engine oil delivery pipe for thermostat to section	Screw	9 Nm	B
Fastening engine oil pipes leading from tank	Screw	10 Nm	B
Fastening engine oil pipe to water/oil pump	Screw	10 Nm	B



- With the water/oil pump on the workbench, check that there are no leaks from the drain orifices.
- When refitting the same oil/water pump, check the state of wear of the splines on the shaft (1).
- Apply a thin film of **MOLYKOTE BR2** on the shaft (1).

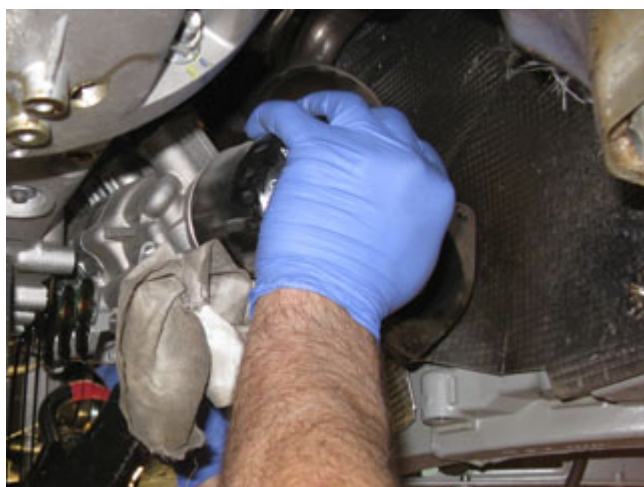


- Apply a thin film of **SHELL ALVANIA RL3** to the O-ring seats before installing the O-rings.
-  Always replace the O-rings with new components.
- Fit the O-rings (2) on the water/oil pump inlets.



Always replace the O-rings with new components.

- Replace the O-ring (11) on the lower crankcase shell joint and grease the relative seat with **SHELL ALVANIA RL3** before reassembly.



When installing the water/oil pump, take particular care not to pinch the new O-rings and that they remain in their relative seats.

- Fit the complete water/oil pump on the crankcase, inserting the splined shaft into the pump drive shaft.

- Hand-tighten all the indicated screws fastening the water/oil pump.
- The screws must be tightened according to the following procedure and sequence:

 Pre-tighten all the screws in the sequence given.

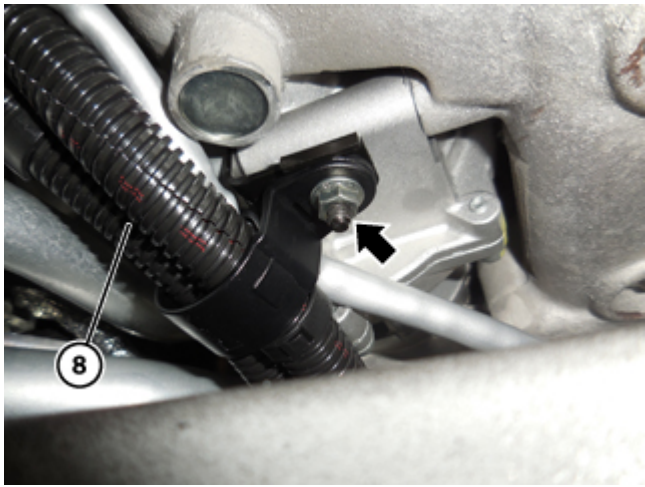
 Tighten the screws in the same sequence.



- Tighten the indicated screw.

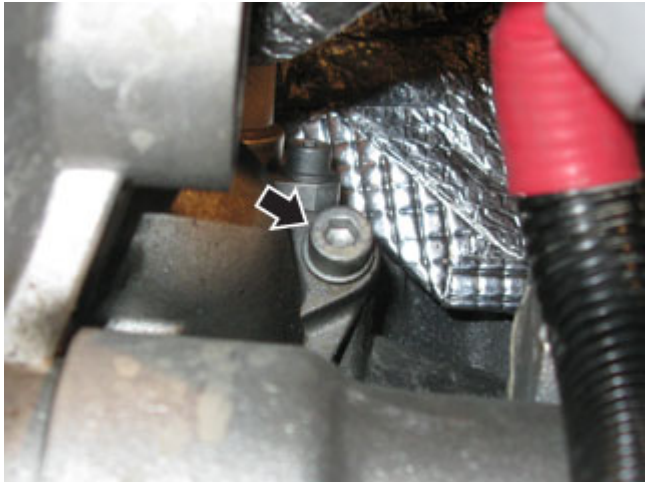


Tightening torque	Nm	Class
Screw (pretightening)	5 Nm	0
Screw	25 Nm	B



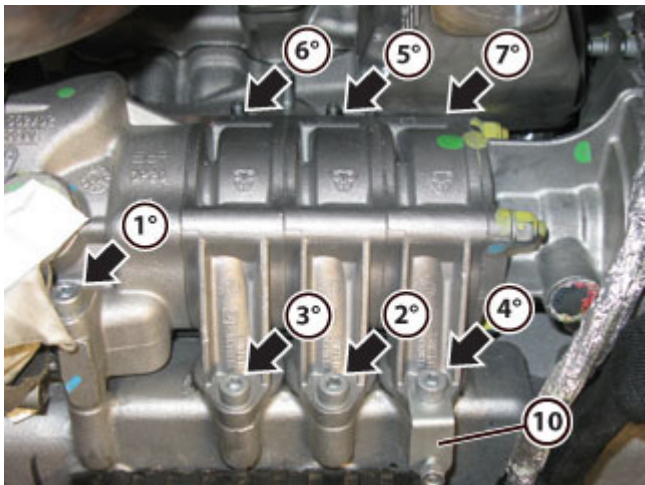
- Tighten the indicated screw fastening the engine services cable (8).

		
Tightening torque	Nm	Class
Screw (pretightening)	5 Nm	0
Screw	25 Nm	B



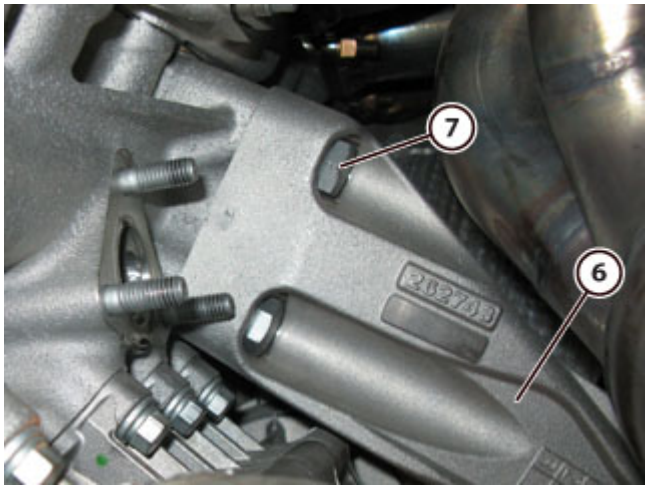
- Tighten the indicated screw.

		
Tightening torque	Nm	Class
Screw (pretightening)	5 Nm	0
Screw	25 Nm	B



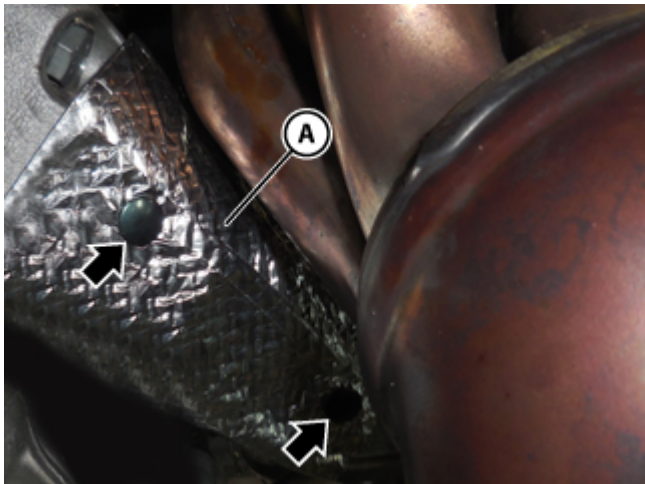
- Fit the bracket (10) then tighten the indicated screws in the sequence given.

		
Tightening torque	Nm	Class
Screw (pretightening)	5 Nm	0
Screw	25 Nm	B

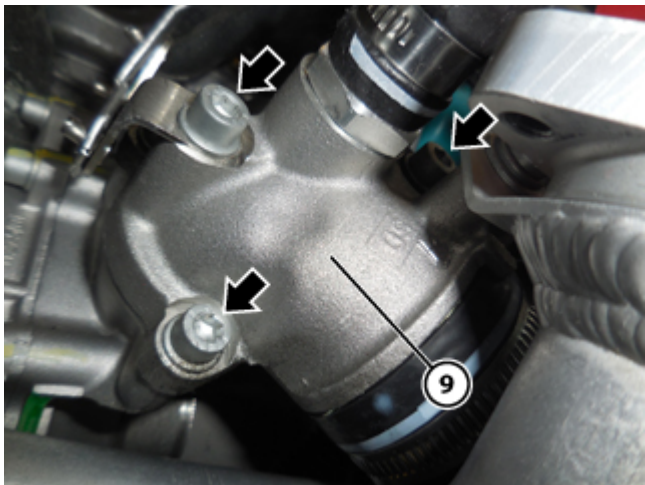


- Fit the engine mounting strut **(6)** in the relative seat and fasten by tightening the four screws **(7)**.

		
Tightening torque	Nm	Class
Screw	45 Nm	A

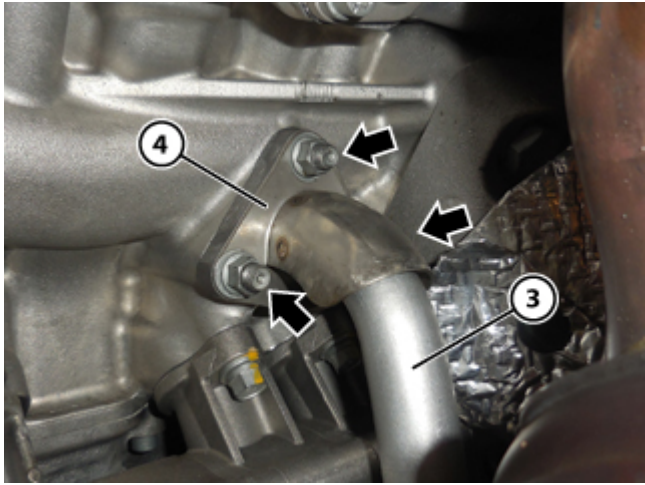


- Fit and fasten the heat shield **(A)**, fastening the indicated clips.



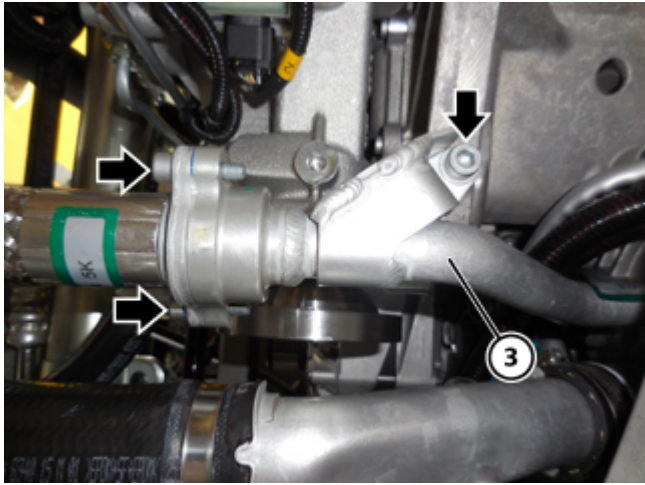
- Apply **SEGO** to the O-ring seat and replace the O-ring.
- Fit the hose union **(9)** on the pump and tighten the indicated screws.

		
Tightening torque	Nm	Class
Screw	10 Nm	B



- Apply **SEGO** to the O-ring seat and replace the O-ring.
- Fit the engine oil pipe **(3)** on the relative studs.
- Fit the shield **(4)** and tighten the indicated nuts.

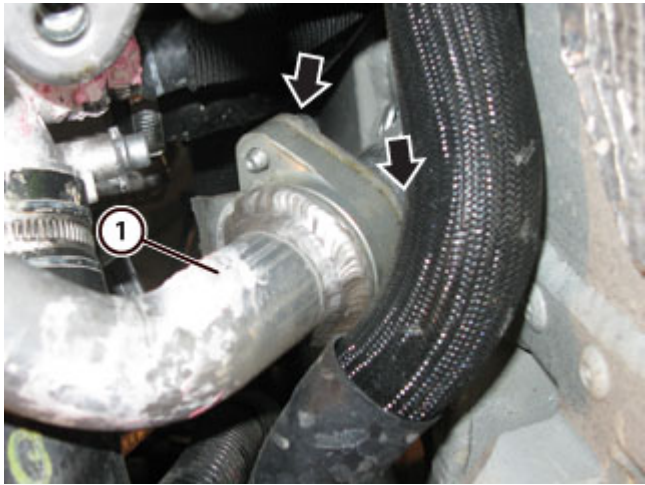
		
Tightening torque	Nm	Class
Nut	20 Nm	B



- Tighten the screws, after applying **AREXONS SYSTEM 52A43 - MEDIUM STRENGTH THREAD LOCK** to the threads.

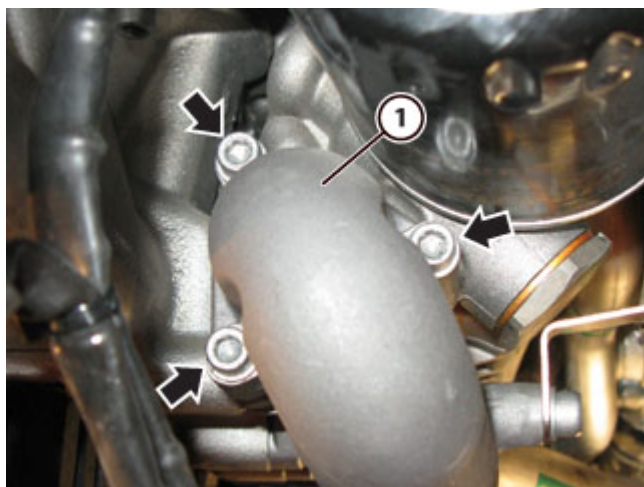
		
Tightening torque	Nm	Class
Screw	9 Nm	B

- Fasten the engine oil pipe with the indicated screw.



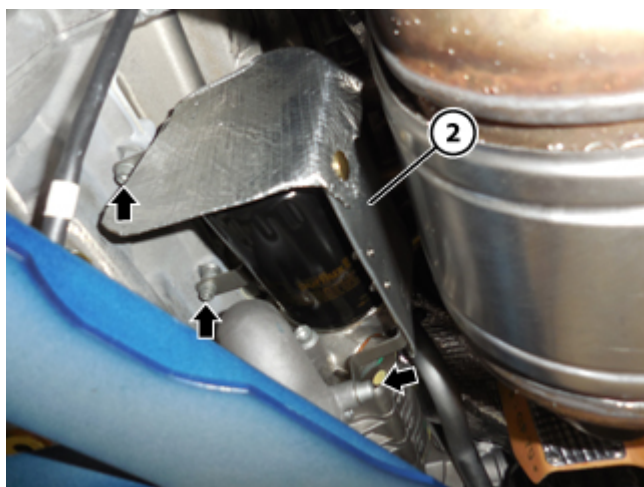
- Apply **SEGO** to the seat on the pipe, replace the O-ring, then connect the engine oil pipe **(1)**, tightening the indicated screws.

		
Tightening torque	Nm	Class
Screw	10 Nm	B



- Apply **SEGO** to the seat on the pump, replace the O-ring, then fasten the engine oil pipe **(1)**, tightening the indicated screws.

		
Tightening torque	Nm	Class
Screw	10 Nm	B



- Fasten the heat shield **(2)** by tightening the indicated screws.

- Refit the catalytic converters ( **B7.03**).
- Refit the right hand catalytic converter.
- Refit the front removable subframe ( **E2.02**).
- Fill the engine lubrication system ( **A3.01**).
- Fill the engine coolant system ( **A3.07**).
- Refit the front wheelhouses ( **E3.05**).
- Refit the front right hand wheelhouse.